



Reading: Exemplar: Maintain efficient process utilization on Linux

Introduction

In this lab, you used the new commands you learned to do some process maintenance on a Linux virtual machine.

This exemplar is a walkthrough of the previous Qwiklab activity, including detailed instructions and solutions. You may use this exemplar if you were unable to complete the lab and/or you need extra guidance in competing lab tasks. You may also refer to this exemplar to prepare for the graded quiz in this module.

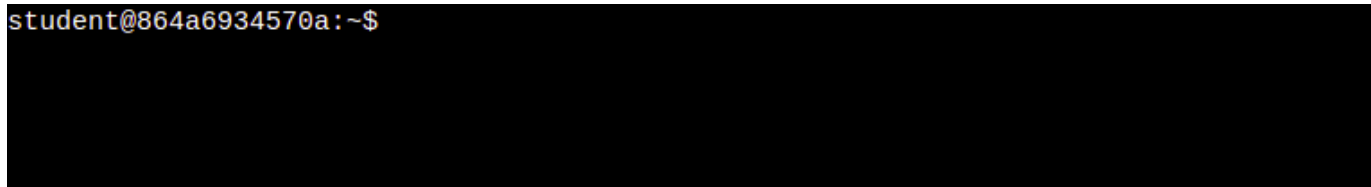
Start the lab

You'll need to start the lab before you can access the materials. To do this, click the green **Start Lab** button at the top of the screen.

A green rectangular button with the text "Start Lab" in white.

After you click the **Start Lab** button, you will see a shell, where you will be performing further steps in the lab. You should have a shell like this:

```
student@864a6934570a:~$
```

A screenshot of a terminal window with a black background. The prompt "student@864a6934570a:~\$" is visible at the top in a light blue font. The rest of the terminal area is empty.

Terminate a specific process

The **ps -aux** command allows you to list all currently running processes on a Linux machine. However, the list of processes is often super long, which makes finding a specific process pretty tough. To filter the processes you're interested in, you can pipe the output of **ps** through **grep**.

There are two "malicious" processes currently running on your machine, called **totally_not_malicious**. You can run **ps** and **grep** to find them, using this command:

```
ps -aux | grep "totally_not_malicious"
```

You should see output similar to this. The top two lines are the two processes, while the last line is the **grep** process you just used to search for them. Check out the three-digit numbers on the left of each of the rows (in bold below); these are the process IDs.

OUTPUT:

```
root      315  0.0  0.0  7232   520 ?      S   15:29   0:00 sudo nohup/home/totally_not_malicious
root      320  0.0  0.1  3652   892 ?      S   15:29   0:00 bash /home/totally_not_malicious
student   5431  0.0  0.1   3084   880 pts/0   S+  15:29   0:00 grep totally_not_malicious
```

To stop a process, you can use the **kill** command. You need to use **sudo** to have permission to stop them. You also need to specify the ID of the process, which will likely be different on your machine than what's shown above (the ID is highlighted in the above output):

```
sudo kill [PROCESS ID]
```

After killing the processes, you can verify that they're no longer running by running the original command again:

```
ps -aux | grep "totally_not_malicious"
```

OUTPUT:

```
student   5783  0.0  0.1   3084   884 pts/0   S+  16:15   0:00 grep totally_not_malicious
```

Terminate multiple processes

There are also multiple processes running on your computer containing the word "razzle". You can find them in the same way that you found the previous process using **ps**. Because **grep** doesn't look for full matches, it can be used to find any process that contains a specific substring:

```
ps -aux | grep "razzle"
```

The below shows all six processes that contain the word "razzle". (Again, you can ignore the last process because it's the process running **grep**.)

OUTPUT:

```
root      328  0.0  0.6  7572  3616 ?      S   06:33   0:00 sudo nohup bash /home/razzle_dazzle
root      329  0.0  0.5  7572  3480 ?      S   06:33   0:00 sudo nohup bash /home/my_cat_razzle
```

```

root    330  0.0  0.5  7572  3520  ?    S   06:33  0:00  sudo nohup bash /home/razzles
root    331  0.0  0.4  3648  2544  ?    S   06:33  0:00  bash /home/my_cat_razzle
root    332  0.0  0.4  3648  2624  ?    S   06:33  0:00  bash /home/razzles
root    333  0.0  0.4  3648  2680  ?    S   06:33  0:00  bash /home/razzle_dazzle
student 773  0.0  0.1  3080   888 pts/0 S+  06:37  0:00  grep razzle

```

To kill each of the processes, you can use the same **kill** command as before, substituting in each process ID:

```
sudo kill [PROCESS ID]
```

To verify that the processes were successfully stopped, you can use the same command you used to find them in the first place:

```
ps -aux | grep "razzle"
```

You should only see the process for the grep command, indicating that the other processes are no longer running:

OUTPUT:

```
student 870  0.0  0.1  3080   880 pts/0 S+  06:38  0:00 grep razzle
```

Conclusion

Woohoo! You've successfully used **ps** to find processes on Linux, and used **kill** to end them. These are common Linux commands, so we recommend you practice until you feel comfortable using them.
